Arista Access Node BAS-ACC-54 Optical Transport Failure SOP

Node: BAS-ACC-54 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_6606

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve Optical Transport Failures on the Arista Access Node BAS-ACC-54, located at Basavanagudi POP. The failure is characterized by the Alarm Code ALARM_6606 and is classified as Critical. The root cause of this failure domain is attributed to the degradation of SFP/XFP transceivers or fiber cuts, which result in Rx/Tx power loss. The Optical Transport Network (OTN) protocol relies on the transmission and reception of optical signals over fiber-optic cables to maintain connectivity between nodes. In the event of a failure, the OTN protocol may not be able to maintain the required signal quality, leading to a Critical alarm. As the Access Node is responsible for providing connectivity to the network, a failure in the Optical Transport domain can have significant impact on network availability and performance.

Pre-requisite Checks

- Requirement: Verify that the Basavanagudi POP is accessible and that the remote hands are dispatched to the site for assistance.

- Requirement: Check the status of the redundant link to ensure that it is operational and can assume traffic in case of a failure.

- Requirement: Confirm that the Arista Access Node BAS-ACC-54 is powered on and in a stable state.

Potential Root Causes

Cause	Probability
SFP/XFP transceiver degradation	High
Fiber cuts	Medium
Rx/Tx power loss due to OTN protocol misconfiguration	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces transceiver detail, Expected Output: Transceiver information, including temperature, voltage, and laser bias current, Description: This command provides detailed information about the transceiver, including its temperature, voltage, and laser bias current, which can help identify any issues related to the SFP/XFP.

Command: show interfaces optical, Expected Output: Optical signal quality information, including optical power and signal-to-noise ratio (SNR), Description: This command provides information about the optical signal quality, including optical power and signal-to-noise ratio (SNR), which can help identify any issues related to the fiber
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cuts or Rx/Tx power loss.

Command: show otu1 otu2 parameters, Expected Output: OTN protocol
parameters, including OTU1 and OTU2 settings, Description: This
command provides information about the OTN protocol parameters,
including OTU1 and OTU2 settings, which can help identify any
misconfigurations that may be contributing to the failure.
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Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP transceiver, Command: replace transceiver, Description: Replace the faulty transceiver with a new one to restore connectivity.

Step 2: Step: Verify the fiber integrity, Command: show interfaces optical, Description: Verify that the fiber is not cut or damaged and that the optical signal quality is within acceptable limits.

Step 3: Step: Reconfigure the OTN protocol parameters, Command: config otu1 otu2 parameters, Description: Reconfigure the OTN protocol parameters to ensure that they are set correctly and that the failure is resolved.

Post-Check Verification

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Command: show interfaces transceiver detail, Expected Output:
Transceiver information, including temperature, voltage, and laser
bias current, Description: Verify that the transceiver is functioning
correctly and that the temperature, voltage, and laser bias current
are within acceptable limits.

Command: show interfaces optical, Expected Output: Optical signal
quality information, including optical power and signal-to-noise ratio
(SNR), Description: Verify that the optical signal quality is within
acceptable limits and that the fiber is not cut or damaged.

Command: show otu1 otu2 parameters, Expected Output: OTN protocol
parameters, including OTU1 and OTU2 settings, Description: Verify that
the OTN protocol parameters are set correctly and that the failure is
resolved.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver, Command: replace transceiver, Description: Restore the original transceiver to its previous state in case the replacement fails to resolve the issue.

- Step: Revert the OTN protocol parameters to their original settings, Command: config otu1 otu2 parameters, Description: Revert the OTN protocol parameters to their original settings in case the reconfiguration fails to resolve the issue.